OsdagBridge

Open Source Software for Steel Girder Bridge Design

Design Report

Project Name	OsdagBridge Demo Bridge
Project Location	Mumbai, India
Author / Designer	Local Validation
Reviewer	
Organization	IIT Bombay
Client Name and Organization	FOSSEE
Job Number	DEMO-LL-001
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Executive Summary

This section provides a concise summary of the bridge design, key inputs, governing loads, and final design outcomes.

Project Overview

Bridge Type	
Design Standard	IRC 5, IRC 6, IRC 22, IRC 24, IS 800
Span	30 m
Carriageway Width	
No. of Girders	4
Girder Spacing	
Deck Thickness	
Overall Design Status	Pass
Governing Check	
Overall Utilization Ratio (max)	0.82 (Girder)

[PLACEHOLDER: Figure 1 – Overall Bridge Plan]

Figure 1 – Overall Bridge Plan

[PLACEHOLDER: Figure 2 – Typical Cross-Section (with girder, deck, barriers, footpath)]

Figure 2 – Typical Cross-Section (with girder, deck, barriers, footpath)

[PLACEHOLDER: Figure 3 – 3D View of Bridge Superstructure]

Figure 3 – 3D View of Bridge Superstructure

Table 1 – Final Bridge Geometry (after optimization)

Member ID				
Section Designation				
Governing Check				
Utilization Ratio	0.82 (Girder)	0.82 (Girder)	0.82 (Girder)	0.82 (Girder)

Note: Utilization ratio (UR) = demand / capacity. A value < 1.0 *indicates a passing check.*

Key Design Outcomes Summary

Girder design pass
Cross bracing design pass
End Diaphragm design pass
Deck design pass

Design Assumptions and Limitations

- Additional inputs not provided by the user were assumed by software per IRC/IS code defaults or practical consideration.
- Grillage analysis was performed using OSPGrillage assuming simply supported I-girders.
- Substructure and foundation design are not included in this report.
- Splice connections and bearings are not designed in this version.

1 Project Information

This section records all project metadata as entered by the designer.

1.1 Project and Design Team Details

Project Name	OsdagBridge Demo Bridge
Project Location	Mumbai, India
Designer	Local Validation
Reviewer	
Organization	IIT Bombay
Client	FOSSEE
Software Version	OsdagBridge

1.2 Applicable Codes and Standards

- Indian Roads Congress (IRC) 5: General Features of Design
- Indian Roads Congress (IRC) 6: Loads and Load Combinations
- Indian Roads Congress (IRC) 22: Composite Construction (Limit State Design)
- Indian Roads Congress (IRC) 24: Steel Road Bridges (Limit State Method)
- Indian Roads Congress (IRC) 112: Concrete Road Bridges (deck design)
- Indian Roads Congress Special Publication (IRC SP) 114: Seismic Design of Road Bridges
- Indian Standard (IS) 800: General Construction in Steel
- Indian Standard (IS) 2062: Hot Rolled Structural Steel Specification
- Indian Standard (IS) 6006: Steel Bearings

2 Input Parameters

This section documents all inputs provided to OsdagBridge. User-provided inputs are clearly distinguished from software-assumed defaults. Where the user did not supply a value, the software has applied the IRC/IS code default or an empirical guideline; these are annotated with an asterisk (*).

2.1 Basic Inputs (User-Defined)

Note: These inputs are mandatory and were provided by the user.

Table 2.1: Project Location

parameter	**value**
Project Location	Mumbai, India
Latitude / Longitude	,
Seismic Zone (IRC 6)	III
Basic Wind Speed (IRC 6)	39 m/s
Shade Temp. Max / Min (IRC 6)	45 °C / 5 °C

Table 2.2: Bridge Geometry

parameter	**value**
Type of Structure	
Span (m)	30 m
Carriageway Width (m)	
Include Median	
Footpath	Both Sides
Skew Angle (degrees)	(IRC 24 Cl. 504.8 limit: ±15°)

Table 2.3: Material Selection

parameter	**value**
Girder Steel Grade (IS 2062)	
Cross Bracing Steel Grade	
End Diaphragm Steel Grade	
Concrete Deck Grade (IRC 22)	

* *Software default value*

2.2 Additional Inputs

Where the user has modified additional inputs, those values are reported here. Where no modification was made, the software default is shown.

Table 2.4: Typical Section Details

parameter	value
Overall Bridge Width (m)	
No. of Girders	4
Girder Spacing (m)	
Deck Overhang Width (m)	
Deck Thickness (mm)	
Footpath Width (m)	(IRC 5 Cl. 104.3.6 min: 1.5 m)
No. of Traffic Lanes	2 (per IRC 5 Cl. 104.3.1)

Table 2.5: Components Details

parameter	value
Crash Barrier Type	
Crash Barrier Load (kN/m)	5.0
Railing Type	
Railing Load (kN/m)	0.75
Wearing Course Material	Bituminous
Wearing Course Thickness (mm)	65 mm

Table 2.6: Girder General Information

Girder	Member ID	Design Mode	Girder Type	Girder Symmetry
		Optimized		
		Optimized		
		Optimized		
		Optimized		

Table 2.7: Girder Section Dimensions

Girder	Total Depth, D (mm)	Web, t_w (mm)	Top Flange (b_{tf}, t_{tf}) mm	Bottom Flange (b_{bf}, t_{bf}) mm
			,	,
			,	,
			,	,
			,	,

Table 2.8: Girder Restraint and Stiffener Details

Girder	Torsional / Warping Restraint	Web Philosophy	Intermediate Stiffeners	Longitudinal Stiffeners	Bearing Stiffener
	,		; Spacing: ; Thickness:		No.: ; Spacing: ; Thickness:
	,		; Spacing: ; Thickness:		No.: ; Spacing: ; Thickness:
	,		; Spacing: ; Thickness:		No.: ; Spacing: ; Thickness:
	,		; Spacing: ; Thickness:		No.: ; Spacing: ; Thickness:

Table 2.9: Member Properties: Cross Bracing Details

Location	Member IDs	Type of Bracing	Bracing Section	Spacing (m)

Table 2.10: Member Properties: End Diaphragm Details

Location	Member IDs	Type of Bracing	Bracing Section

Table 2.11: Shear Connector Details

parameter	value
Stud Diameter (mm)	
Stud Height (mm)	
Stud f_y (MPa)	
Stud f_u (MPa)	
No. of Studs per Section	

Note: All values are per IRC 22 Table 1 unless user-modified.

Table 2.12: Partial Safety Factors

parameter	value
γ_{M0} (Yielding / Buckling)	
γ_{M1} (Ultimate Stress)	
γ_{C} (Concrete, Basic)	
γ_{s} (Reinforcement)	
γ_{v} (Shear Connectors)	
γ_{fft} (Fatigue Load)	
γ_{Mft} (Fatigue Strength)	

3 Loads and Load Combinations

This section summarizes all loads applied to the bridge and the load combinations considered for analysis and design.

Table 3.1: Dead Load – Self Weight

parameter	value
Steel Self-Weight Applied	78.5 kN/m^3
Concrete Deck Weight	25 kN/m^3
Self-Weight Factor	1.05

Table 3.2: Dead Load for Surfacing (DW)

parameter	value
Wearing Course Load	Bituminous x 65
Additional SIDL (Crash Barrier)	5.0 kN/m per barrier
Railing Load	0.75 kN/m*

Table 3.3: Live Loads (LL)

parameter	value
Vehicles Considered	Class A, Class 70R (Wheeled), Class SV
Impact Factor (IRC 6)	Class A: 1.207, Class AA/70R: 1.100
Braking Load (IRC 6)	N/A
Footpath Live Load (if applicable)	5.0 kN/m^2

** Software default value*

Table 3.4: Wind Load (WL) — per IRC 6

parameter	value
Basic Wind Speed, Vb	39 m/s [from Project Location]
Terrain Type	
Average Exposed Height, H (m)	
Hourly Mean Wind Speed, Vz	
Hourly Wind Pressure, Pz	
Transverse Wind Force	
Longitudinal Wind Force	
Vertical Wind Force	

Table 3.5: Earthquake Load (EL) — per IRC 6

parameter	value
Seismic Zone	III [from Project Location]
Zone Factor, Z	
Importance Factor, I	
Type of Soil	
Sa/g	
Horizontal Seismic Coefficient, Ah	
Vertical Seismic Coefficient, Av	
Horizontal Seismic Force (longitudinal)	kN
Horizontal Seismic Force (transverse)	kN

Table 3.6: Temperature Load (TL) — per IRC 6

parameter	value
Maximum Shade Temperature	45 °C
Minimum Shade Temperature	5 °C
Effective Bridge Temp. Range	to °C
Temperature Rise / Fall for Design	+ °C / − °C

Table 3.7: Load Combinations

Combination ID	Load Cases
ULS-01	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.50) + WL(0.90) + TL(0.90)
ULS-02	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(1.50) + TL(0.90)
ULS-03	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(0.90) + TL(1.50)
ULS-04	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + VC(1.00)
ULS-05	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + BI(1.00)
ULS-06	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + FB(1.00)
ULS-07	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(1.50)
ULS-08	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(0.75)
SLS-01	DL(1.00) + SIDL(1.20/1.00) + LL(1.00) + WL(0.60) + TL(0.60)
SLS-02	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(1.00) + TL(0.60)
SLS-03	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.60) + TL(1.00)
SLS-04	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.50) + TL(0.50)
SLS-05	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.60) + TL(0.50)

SLS-06	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.50) + TL(0.60)
SLS-07	DL(1.00) + SIDL(1.20/1.00) + LL(0.00) + WL(0.00) + TL(0.50)

Note: All IRC 6 load combinations are auto-generated by OsdagBridge. User-defined custom combinations, if any, are appended.

4 Analysis Results

A grillage model was used for structural analysis. The deck is idealized as a grid of elastic beam elements — longitudinal members represent the composite steel girders with effective slab, and transverse members represent the slab or cross frames. This section summarizes the critical output from that analysis.

Table 4.1: Summary of Maximum Demands

Load Case/ Comb.	Bending Moment			Shear Force			Reaction at Supports	
	Max (kNm)	Girder	Loc. (m)	Max (kN)	Girder	Loc. (m)	Left	Right
—	—	—	—	—	—	—	—	—

Table 4.2: Reactions at Supports

Load Case	Left Support (kN)	Right Support (kN)

Table 4.3: Deflection Summary (Live Load & Total Load)

parameter	value
Deflection due to Live Load, δ_{LL}	—
Allowable Live Load Deflection (Δ_{allow})	L/800 = 37.5 mm
Live Load Deflection Check Status	—
Deflection due to Total Load, δ_{total}	—
Allowable Total Deflection (Δ_{allow})	L/600 = 50.0 mm
Total Load Deflection Check Status	—

[PLACEHOLDER: Bending Moment Envelope (Envelope ULS): Max/min BM along span. X-axis: distance from left support (m). Y-axis: Bending Moment (kN-m).]

[PLACEHOL
X-axis: distan

[PLACEHOLDER: Vertical Deflection D_y (1.0 LL): Maximum deflection along span. Load Case: 1.0 LL, Combination: D_y. Nodes shown. Isometric view.]

5 Design Checks

This section presents all structural design checks performed by OsdagBridge. For each member, the demand from the governing load combination, the code-based capacity, and the utilization ratio are tabulated. All checks reference IS 800:2007 and IRC 22:2014 unless stated otherwise.

5.1 Plate Girder Design

Table 5.1: Girder Section Properties (Final Optimized / User-selected)

Girder	Property	Value
	Depth, D (mm)	
	Top Flange Width, b_f (mm)	
	Bottom Flange Width, b_f (mm)	
	Top Flange Thickness, t_f (mm)	
	Bottom Flange Thickness, t_f (mm)	
	Web Thickness, t_w (mm)	
	Gross Area, A (cm^2)	
	Moment of Inertia, I_z (cm^4)	
	Elastic Section Modulus, Z_{ez} (cm^3)	
	Plastic Section Modulus, Z_{pz} (cm^3)	
	Effective Slab Width, b_{eff} (mm)	
	Transformed Composite I_z (cm^4)	
	Depth to Plastic Neutral Axis (mm)	
	Depth, D (mm)	
	Top Flange Width, b_f (mm)	
	Bottom Flange Width, b_f (mm)	
	Top Flange Thickness, t_f (mm)	
	Bottom Flange Thickness, t_f (mm)	
	Web Thickness, t_w (mm)	

	Gross Area, A (cm^2)	
	Moment of Inertia, I_z (cm^4)	
	Elastic Section Modulus, Z_{ez} (cm^3)	
	Plastic Section Modulus, Z_{pz} (cm^3)	
	Effective Slab Width, b_{eff} (mm)	
	Transformed Composite I_z (cm^4)	
	Depth to Plastic Neutral Axis (mm)	
	Depth, D (mm)	
	Top Flange Width, b_f (mm)	
	Bottom Flange Width, b_f (mm)	
	Top Flange Thickness, t_f (mm)	
	Bottom Flange Thickness, t_f (mm)	
	Web Thickness, t_w (mm)	
	Gross Area, A (cm^2)	
	Moment of Inertia, I_z (cm^4)	
	Elastic Section Modulus, Z_{ez} (cm^3)	
	Plastic Section Modulus, Z_{pz} (cm^3)	
	Effective Slab Width, b_{eff} (mm)	
	Transformed Composite I_z (cm^4)	
	Depth to Plastic Neutral Axis (mm)	
	Depth, D (mm)	
	Top Flange Width, b_f (mm)	
	Bottom Flange Width, b_f (mm)	
	Top Flange Thickness, t_f (mm)	
	Bottom Flange Thickness, t_f (mm)	
	Web Thickness, t_w (mm)	
	Gross Area, A (cm^2)	

	Moment of Inertia, I_z (cm^4)	
	Elastic Section Modulus, Z_{ez} (cm^3)	
	Plastic Section Modulus, Z_{pz} (cm^3)	
	Effective Slab Width, b_{eff} (mm)	
	Transformed Composite I_z (cm^4)	
	Depth to Plastic Neutral Axis (mm)	

Table 5.2: Girder Section Classification

	Element	**Slenderness Ratio**	**Class Limit**	**Classification**
	Top Flange			
	Web			
	Overall Section	—	—	
	Top Flange			
	Web			
	Overall Section	—	—	
	Top Flange			
	Web			
	Overall Section	—	—	
	Top Flange			
	Web			
	Overall Section	—	—	

Note: IS 800:2007 Table 2

Table 5.3: Moment Capacity Check

	Parameter	Formula	Value	Status
	Applied Moment, M_u	Governing LC (ULS)		—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1		—
	Utilization Ratio, M_u/M_d	M_u/M_d		—
	Applied Moment, M_u	Governing LC (ULS)		—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1		—
	Utilization Ratio, M_u/M_d	M_u/M_d		—
	Applied Moment, M_u	Governing LC (ULS)		—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1		—
	Utilization Ratio, M_u/M_d	M_u/M_d		—
	Applied Moment, M_u	Governing LC (ULS)		—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1		—
	Utilization Ratio, M_u/M_d	M_u/M_d		—

Note: IRC 22 Cl. 603.3.1, IS 800 Cl. 8.2.1

Table 5.4: Shear Capacity Check

	Parameter	Formula	Value	Status
	Applied Shear, V_u	Governing LC (ULS)		—
	Shear Area, A_v	$d_w \times t_w$		—
	Panel Aspect Ratio, c/d	—		—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2		—

	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$		—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2		—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$		—
	Utilization Ratio, V_u/V_d	V_u/V_d		—
	Applied Shear, V_u	Governing LC (ULS)		—
	Shear Area, A_v	$d_w \times t_w$		—
	Panel Aspect Ratio, c/d	—		—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2		—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$		—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2		—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$		—
	Utilization Ratio, V_u/V_d	V_u/V_d		—
	Applied Shear, V_u	Governing LC (ULS)		—
	Shear Area, A_v	$d_w \times t_w$		—
	Panel Aspect Ratio, c/d	—		—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2		—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$		—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2		—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$		—

	Utilization Ratio, V_u/V_d	V_u/V_d		—
	Applied Shear, V_u	Governing LC (ULS)		—
	Shear Area, A_v	$d_w \times t_w$		—
	Panel Aspect Ratio, c/d	—		—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2		—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$		—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2		—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$		—
	Utilization Ratio, V_u/V_d	V_u/V_d		—

Note: IS 800 Cl. 8.4, IRC 22 Cl. 603.3.3.2

Table 5.5: Interaction Checks (M-V and M-N)

	Check	Condition	Value	Status
	High Shear Condition?	$V_u > 0.6\,V_d$		—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
	Interaction Check: $M_u \leq M_{dv}$	—		—
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A
	High Shear Condition?	$V_u > 0.6\,V_d$		—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
	Interaction Check: $M_u \leq M_{dv}$	—		—
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A
	High Shear Condition?	$V_u > 0.6\,V_d$		—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
	Interaction Check: $M_u \leq M_{dv}$	—		—
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A
	High Shear Condition?	$V_u > 0.6\,V_d$		—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3		—
	Interaction Check: $M_u \leq M_{dv}$	—		—
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	N/A	N/A	N/A

Note: IRC 22 Cl. 603.3.3.3

Table 5.6: Lateral Torsional Buckling Check – Construction Stage

	Parameter	Formula	Value	Status
	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1		—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$		—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2		—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$		—
	$M_u \leq M_b$	M_u/M_b		—
	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1		—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$		—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2		—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$		—
	$M_u \leq M_b$	M_u/M_b		—
	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1		—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$		—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2		—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$		—
	$M_u \leq M_b$	M_u/M_b		—
	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1		—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$		—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2		—

	LTB Resistance, M_b	$\chi_{LT} M_p/\gamma_{m0}$		—
	$M_u \leq M_b$	M_u/M_b		—

Note: IRC 22 Cl. 603.3.3.1, IS 800 Cl. 8.2.2

Table 5.7: Stiffener Design Summary

	Shear Buckling Design Method	
	Intermediate Stiffener Thickness (mm)	
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	
	No. of End Panel Stiffeners	
	Longitudinal Stiffeners	
	Shear Buckling Design Method	
	Intermediate Stiffener Thickness (mm)	
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	
	No. of End Panel Stiffeners	
	Longitudinal Stiffeners	
	Shear Buckling Design Method	
	Intermediate Stiffener Thickness (mm)	
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	
	No. of End Panel Stiffeners	
	Longitudinal Stiffeners	
	Shear Buckling Design Method	

	Intermediate Stiffener Thickness (mm)	
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	
	No. of End Panel Stiffeners	
	Longitudinal Stiffeners	

Table 5.8: End Panel Stiffener Checks

	Check	Required	Provided	Status
	Web Buckling Resistance	N/A	N/A	N/A
	Local Crushing Resistance	N/A	N/A	N/A
	Bearing Capacity	N/A	N/A	N/A
	Column Buckling Resistance	N/A	N/A	N/A
	Web Buckling Resistance	N/A	N/A	N/A
	Local Crushing Resistance	N/A	N/A	N/A
	Bearing Capacity	N/A	N/A	N/A
	Column Buckling Resistance	N/A	N/A	N/A
	Web Buckling Resistance	N/A	N/A	N/A
	Local Crushing Resistance	N/A	N/A	N/A
	Bearing Capacity	N/A	N/A	N/A
	Column Buckling Resistance	N/A	N/A	N/A
	Web Buckling Resistance	N/A	N/A	N/A

	Local Crushing Resistance	N/A	N/A	N/A
	Bearing Capacity	N/A	N/A	N/A
	Column Buckling Resistance	N/A	N/A	N/A

Note: IS 800 Cl. 8.4.2.2

Table 5.9: Serviceability – Deflection Checks

	Check	**Allowable**	**Actual**	**Status**
	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm		—
	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm		—
	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm		—
	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm		—
	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm		—
	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm		—
	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm		—
	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm		—

Note: IRC 22 Cl. 604.3.2

Table 5.10: Serviceability – Maximum Stress Limitation

	Element	Allowable Stress	Actual Stress	Status
	Structural Steel ($0.9\, f_y$)			—
	Structural Steel ($0.9\, f_y$)			—
	Structural Steel ($0.9\, f_y$)			—
	Structural Steel ($0.9\, f_y$)			—

Table 5.11: Serviceability – Fatigue Assessment

	Stress Range, $\Delta\sigma$ (MPa)	Fatigue Limit, f_{fd} (MPa)	Utilization Ratio	Status
				—
				—
				—
				—

Note: IRC 22 Cl. 605 — governing of normal and shear fatigue (worst by DCR). Capacity reduction factor μ_r applied where plate thickness ¿ 25 mm.

Table 5.12: Girder Design Summary (DCR / Utilization Ratio)

Girder	Controlling LC / Combination	Controlling Check	Demand	Capacity	UR	Status
	Basic ULS: 1.35DL + 1.5LL		2450.00 kNm	3000.00 kNm	0.820	—

Note: UR = Demand / Capacity. A value ≤ 1.0 indicates a passing check. The controlling check is the criterion with the highest UR for each girder, with the real load case/combination that drives it.

Table 5.13: Shear Connector Capacity

Parameter	Formula	Value	Reference
Design Resistance, Q_u	$Q_u = \min(Q_{u,s},\, Q_{u,c})$ $Q_{u,s} = \dfrac{0.8\, f_u\, (\pi d^2/4)}{\gamma_v}$ $Q_{u,c} = \dfrac{0.29\, \alpha\, d^2 \sqrt{f_{ck}\, E_{cm}}}{\gamma_v}$		IRC 22 Cl. 606.3.1 (Eq. 6.1)
Fatigue Shear Resistance, Q_r	IRC 22 Table 8 ($\phi d,\ N_{sc}$)		IRC 22 Cl. 606.3.2 (Table 8)

Table 5.14: Shear Connector Spacing

Criterion	Governing Spacing	Actual Spacing Provided	Status
ULS Shear (SL1)			—
Full Composite (SL2)			—
SLS Fatigue (SR)			—
Max Spacing Limit (IRC 22)			—

Note: IRC 22 Cl. 606.4, 606.9. Governing spacing $= \min(S_{L1}, S_{L2}, S_R)$.

Table 5.15: Transverse Shear and Detailing Checks

Check	Value	Status
Longitudinal Shear per unit length, V_L		—
Transverse Shear Capacity of Slab, V_{Rd}		—
Transverse Shear Check	$V_L/V_{Rd} =$ —	—
Min. Transverse Reinforcement, $A_{st,min}$	Required , Provided 0.00 cm^2/m	—
Stud Diameter $\leq 2\,t_f$	$d = \leq 2t_f =$	—
Stud Edge Distance	Provided (req. $\geq$)	—

Note: IRC 22 Cl. 606.6, 606.10.

5.2 Deck Slab Design

The reinforced concrete deck slab is designed per IRC 112:2011 (flexure, shear, crack width) and IRC 22:2014 (composite construction). Wheel loads are distributed using Pigeaud's method. The deck is checked for flexure in the transverse and longitudinal directions, punching shear, one-way (beam) shear, crack width, and reinforcement detailing.

Table 5.16: Deck Slab — Loading and Geometry

Effective Span of Deck Slab, l_{eff}	<—> mm (girder spacing, c/c)
Deck Thickness, t_s	mm
Clear Cover (IRC 112 Cl. 15.2)	Top / Bottom mm
Concrete Grade (IRC 112 Cl. 6.4)	($f_{ck} =$ MPa, $f_{ctm} =$ MPa)
Reinforcement Grade (IRC 112 Cl. 6.2)	($f_y =$ <—> MPa)
Dead Load per Unit Area, w_{DL}	<—> kN/m^2 (slab self-weight)
IRC 6 Wheel Load (Class A / 70R)	<—> kN

Tyre Contact Width (IRC 6 Annex A)	<—> mm (transverse)
Impact Factor (IRC 6 Cl. 208.2)	<—>
Governing Live Load Case	<—>

Table 5.17: Deck Slab — Flexure Check: Interior Panel (Pigeaud's Method)

Location	Parameter	Formula / Reference	Value	Status
At Midspan (Sagging)	Transverse BM (DL), $M_{T,DL}$	$w_{DL}\, l_{eff}^2/10$	<—> kN-m/m	—
	Transverse BM (LL), $M_{T,LL}$	Effective width (IRC 112 B3.1)	<—> kN-m/m	—
	Total Design BM, $M_{u,sag}$	<—> DL + <—> LL	<—> kN-m/m	—
	Effective depth, d	$t_s - c_{nom} - \phi/2$	<—> mm	—
	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	<—> kN-m/m	—
At Support (Hogging)	Total Design BM, $M_{u,hog}$	<—> DL + <—> LL (at support)	<—> kN-m/m	—
	Required Top Steel, $A_{st,top}$	$M_u/(0.87\, f_y\, d)$	<—> mm^2/m	—
	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	<—> kN-m/m	—

Note: IRC 112 Cl. 12.2. Distribution (longitudinal) reinforcement designed for 20% of main steel moment (IRC 21 Cl. 305.18).

Table 5.18: Deck Slab — Cantilever Overhang Flexure Check

Parameter	**Formula**	**Value**	**Status**
Overhang Length, l_{oh}	—		—
Crash Barrier Load Moment	IRC 6 Cl. 206.4	<—>	—
Dead Load Moment	$w_{DL}\, l_{oh}^2/2$ + railing	<—>	—
Live Load Moment (eccentric wheel)	Wheel load × arm	<—>	—
Total Hogging Moment, $M_{u,oh}$	<—> DL + <—> (LL + CB)	<—>	—
Moment Capacity (top steel), $M_{Rd,oh}$	IRC 112 Cl. 12.2	<—>	—

Note: IRC 6 Cl. 206.4 crash barrier loads applied at kerb face; IRC 112 Cl. 12.2 flexure.

Table 5.19: Deck Slab — Punching Shear Check (IRC 112 Cl. 10.4.6)

Parameter	**Formula / Reference**	**Value**	**Status**
Design Wheel Load (ULS), V_{Ed}	$\gamma_Q\,(1 + IF)\,P_w$	<—> kN	—
Tyre Contact Area	$a \times b$ (IRC 6 Annex A)	<—> × <—> mm	—
Loaded Area at mid-depth, b_0	$c_1 \times c_2$ (incl. WC dispersion)	<—> × <—> mm	—
Control Perimeter, u_1	$2(c_1 + c_2) + 4\pi d$	<—> mm	—
Punching Shear Stress, v_{Ed}	$V_{Ed}/(u_1\, d)$	<—> MPa	—
Punching Resistance, $v_{Rd,c}$	IRC 112 Eq. 10.1	<—> MPa	—
Punching Shear Check	$v_{Ed} \le v_{Rd,c}$	<—>	—

Note: Punching shear reinforcement not typically required for deck slabs with $d \ge 200$ mm and adequate longitudinal reinforcement.

Table 5.20: Crack Width Check (Deck Slab)

Parameter	Value / Reference
Min. Reinforcement for Crack Control, $A_{s,min}$	<—> mm²/m [IRC 112 Cl. 16.5.1]
Provided Reinforcement (bottom)	ϕ<—> @ <—> mm c/c (<—> mm²/m)
Max. Permissible Crack Width	<—> mm
Calculated Crack Width, w_k (governing)	<—> mm
Crack Width Check	—

Table 5.21: One-Way (Beam) Shear Check (Deck Slab)

Parameter	Formula / Reference	Value	Status
Design Shear per unit width, V_{Ed}	$\gamma_{DL} V_{DL} + \gamma_{LL}(1{+}IF)V_{LL}$	<—> kN/m	—
Effective depth, d	$t_s - c_{nom} - \phi/2$	<—> mm	—
Size factor, k	$1 + \sqrt{200/d} \leq 2.0$	<—>	—
Long. reinforcement ratio, ρ_l	$A_{sl}/(b_w\, d) \leq 0.02$	<—>	—
Shear resistance (no stirrups), $V_{Rd,c}$	$v_{Rd,c}\, b_w\, d$ (Cl. 10.3.2)	<—> kN/m	—
One-Way Shear Check	$V_{Ed} \leq V_{Rd,c}$	<—>	—

Note: IRC 112 Cl. 10.3.2. Shear reinforcement not provided in deck slabs; capacity relies on concrete and main reinforcement.

Table 5.22: Reinforcement Detailing Summary (Deck Slab)

Parameter	Required / Limit	Provided	Status
Main Reinforcement — Bottom (Transverse)			
Required Area, $A_{st,req}$ (mm^2/m)	<—> mm^2/m	<—> mm^2/m	—
Bar Diameter × Spacing	$\phi \geq 10$ mm (IRC 112)	ϕ<—> @ <—> mm c/c	—
Min. Reinforcement $A_{s,min}$ (IRC 112 Cl. 16.3.1)	<—> mm^2/m	<—> mm^2/m	—
Max. Bar Spacing (IRC 112 Cl. 16.3.2)	<—> mm	<—> mm	—
Distribution Reinforcement — Longitudinal			
Required Area, $A_{st,dist}$ (mm^2/m)	≥ 20% of main steel	<—> mm^2/m	—
Top Reinforcement (Support / Cantilever Overhang)			
Required Area, $A_{st,top}$ (mm^2/m)	<—> mm^2/m	<—> mm^2/m	—
Cover and Detailing			
Clear Cover (IRC 112 Cl. 15.2)	≥ <—> mm (Table 14.2)	Top / Bottom mm	—

Note: IRC 112 Cl. 16.3, IS 456 Cl. 26.5. All reinforcement provisions satisfy strength and detailing requirements.

5.3 Cross Bracing Design

Cross bracing between adjacent plate girders provides lateral stability during construction, resists transverse loads (wind, seismic, braking) in service, and prevents lateral torsional buckling of the girders. Members are designed per IS 800:2007 Cl. 7 (compression) and Cl. 6 (tension). Forces are derived from the grillage model under the governing load combination (DL + LL + WL).

Table 5.23: Cross Bracing — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm²)	r_{min} (mm)
Between Girders	Diagonal				

Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

Table 5.24: Cross Bracing — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
Between Girders	Diagonal	C			—

Note: 3. Limit = 250 for compression members, 400 for tension members. $K = 1.0$ for members with both ends pinned.

Table 5.25: Cross Bracing Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.4 End Diaphragm Design

End diaphragms at the supports transfer transverse loads to the bearings, restrain the bottom flanges against lateral displacement, and maintain the girder cross-section geometry during construction and in service. They are designed per IS 800:2007 and IRC 24:2010 Cl. 507.

Table 5.26: End Diaphragm — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm²)	r_{min} (mm)
Between Girders	Diagonal				

Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

Table 5.27: End Diaphragm — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
Between Girders	Diagonal	C			—

Note: 3. Limit = 250 for compression members, 400 for tension members. $K = 1.0$ *for members with both ends pinned.*

Table 5.28: End Diaphragm Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.5 Overall Design Check Summary

Table 5.29: Overall Design Check Summary — All Members

Member / Check	Governing Load Combo	Demand	Capacity	UR
Girder — Moment	Basic ULS: 1.35DL + 1.5LL	2450.00 kNm	3000.00 kNm	0.82
Girder — Shear	—	420.00 kN	690.00 kN	0.61
Girder — LTB (constr.)	—	—	—	—
Girder — Deflection	—	—	—	—
Girder — Stress	—	—	—	—
Girder — Fatigue	—	—	—	—
Transverse Shear (slab)	—	—	—	—
Crack Width (slab)	—	—	—	—
Deck — Flexure (sagging)	—	—	—	—

Deck — Flexure (hogging)	—	—	—	—
Deck — Cantilever Overhang	—	—	—	—
Deck — Punching Shear	—	—	—	—
Deck — One-Way Shear	—	—	—	—
Cross Bracing — Compression	—	—	—	—
Cross Bracing — Tension	—	—	—	—
Cross Bracing — Slenderness	—	—	—	—
End Diaphragm — Moment	Rolled / Welded section — design to be added			
End Diaphragm — Shear	Rolled / Welded section — design to be added			

Note: UR = Demand / Capacity. All values ≤ 1.0 *indicate passing checks. The governing check for each component is highlighted in the individual design check sections above.*

6 Material Take-off & Quantity Summary

Table 7.1 Bill of Materials (Steel, Concrete, and Reinforcement Quantities)

S.N.	Item Description	Volume	Quantity	Total Volume	Weight (MT)	Total Weight (MT)
1	Structural Steel (IS 2062) for Girders	V=A*L	4	1.620	3.150	12.600
2(a)	Cross Bracing - Top Chord	—	12	0.080	0.052	0.624
2(b)	Cross Bracing - Bottom Chord	—	12	0.075	0.049	0.588
2(c)	Cross Bracing - Diagonal Chord	—	24	0.120	0.039	0.936
3	Concrete (M40) for Deck Slab	L*B*t	1	48.500	121.250	121.250
4	Reinforcement Steel (Fe 500)	—	1	0.950	7.450	7.450
5	Shear Stud Connectors	—	800	0.020	0.001	0.800
6	Crash Barrier	—	2	3.200	4.000	8.000

7 Standards & Assumptions

This section provides references to standards used to calibrate the OsdagBridge design modules, and notes the limitations of the current software version.

7.1 Design Standards

The following Indian Road Congress (IRC) codes and Indian Standards (IS) form the basis of all design calculations in this software.

Table 7.1: IRC Codes

Code	Year	Title / Scope
IRC 5	2015	General Features of Design - carriageway widths, kerb, footpath dimensions
IRC 6	2017	Loads and Load Combinations - dead load, live load, impact, wind, temperature, etc.
IRC 22	2015	Composite Construction (LS) - Composite section properties, ULS/SLS design, shear connectors
IRC 24	2010	Steel Road Bridges (LS) - Stiffener design, skew angle limits, diaphragm requirements
IRC 112	2020	Concrete Road Bridges - Deck slab flexure, shear, crack width, reinforcement
IRC SP 114	2018	Seismic Design of Road Bridges

Table 7.2: IS Codes

Code	Year	Scope
IS 800	2007	Steel construction - tension, compression, bending, shear, LTB, stiffeners, combined checks
IS 456	2000	Concrete - simplified stress-block for deck moment capacity
IS 1786	2008	Reinforcement steel properties
IS 1893 (Part 3)	2014	Earthquake resistant design
IS 2062	2011	Structural steel - yield and ultimate strength by grade

7.2 Analysis and Design Assumptions of This Version

Structural Analysis

- All girders are modelled as simply supported; continuous spans are not currently supported.
- A 3D grillage model (OSPGrillage) is used for load distribution. Grillage members carry composite section properties after the construction stage.
- The transverse member forces are computed using an approximate 2D frame analogy. For irregular or skewed geometries, a 3D FEM is recommended.
- Fixed bearing stiffness is modelled as $k = 1{,}000{,}000\,\mathrm{kN/m}$ (virtually rigid); free/expansion bearing as $k = 100\,\mathrm{kN/m}$.
- Construction stage sequence analysis is approximate. Detailed staged analysis should be performed for long-term deflection checks.

Composite Action

- Full shear connection is assumed at ULS with headed stud connectors designed per IRC 22:2015 Cl.606.
- Short-term composite section properties (modular ratio $n = E_s/E_{cm}$) are used for ULS checks.
- Long-term composite section properties (with creep-adjusted modular ratio) are used for SLS deflection and crack-width checks.
- Pre-composite stage: the steel girder alone resists all construction loads prior to concrete gaining strength.

Material Properties (IRC 22:2015 Annex III)

- Steel: $E_s = 200{,}000\,\mathrm{MPa}$, $G_s = 80{,}000\,\mathrm{MPa}$, $\nu = 0.30$, $\alpha = 11.7 \times 10^{-6}/°\mathrm{C}$ (grade-independent).
- Minimum structural concrete grade: M25.
- Default reinforcement grade: Fe500.

Partial Safety Factors (IRC 22:2015 Cl.601.4)

- γ_{m0} (steel yield, ULS) = 1.10
- γ_{m1} (steel ultimate, ULS) = 1.25

- Reinforcement (ULS) = 1.15
- Welds – shop: 1.25; field: 1.50
- Fatigue (γ_{mft}) = 1.35

Loading

- Dead load densities per IRC 6:2017 Cl.203: structural steel 78.5 kN/m^3, concrete 25.0 kN/m^3, bituminous wearing course 24.0 kN/m^3.
- Multi-lane live load reduction factors per IRC 6:2017 Cl.204.4 Table 6A: 1st lane = 1.0, 2nd lane = 0.8, 3rd lane onwards = 0.4.
- Impact factor (dynamic load allowance) computed from span per IRC 6:2017 Cl.208.2/208.3.
- Wind load applied as transverse, longitudinal, and vertical components per IRC 6:2017 Cl.209.3.3–209.3.5.

Serviceability Limits

- Deflection limits (IRC 22:2015 Cl.604.3.2): live load + impact $\leq L/800$; total $\leq L/600$.
- SLS stress limits: concrete $\sigma_c \leq 0.48 f_{ck}$; rebar $\sigma_s \leq 0.80 f_{yk}$; steel $f_e \leq 0.9 f_y$.
- Permissible crack width: $w_k \leq 0.3$ mm (bridge deck, exposure class XS2/XD2 per IRC 112:2020 Cl.12.3.2).

Fatigue

- Reference fatigue life: $N_{sc} = 2 \times 10^6$ cycles (IRC 22:2015 Cl.605).
- Constant stress range is assumed. A thickness correction factor μ_r is applied for plate thickness > 25 mm.

Stiffener Design

- Intermediate transverse stiffener and bearing stiffener design follows IS 800:2007 Cl.8.7.2/8.7.3 and IRC 24:2010 Cl.509.7.2/509.7.3.

7.3 Known Limitations of This Version

- Substructure (piers, pile caps, foundations) and bearing design are not included.
- Splice connection design is not implemented.
- Skew angle > 15 degrees requires independent manual analysis (IRC 24 Cl. 504.8).
- Construction stage sequence analysis is approximate; detailed staged analysis should be performed for long-term deflection checks.
- The grillage analysis assumes simply supported boundary conditions; continuous spans are not currently supported.

References

1. IRC 5 (2024) — *Standard Specifications and Code of Practice for Road Bridges, Section I: General Features of Design.*
2. IRC 6 (2017) — *Standard Specifications and Code of Practice for Road Bridges, Section II: Loads and Load Combinations.*
3. IRC 22 (2014) — *Standard Specifications and Code of Practice for Road Bridges, Section VI: Composite Construction (Limit State Design).*
4. IRC 24 (2010) — *Standard Specifications and Code of Practice for Road Bridges, Section V: Steel Road Bridges (Limit State Method).*
5. IRC 112 (2011) — *Code of Practice for Concrete Road Bridges.*
6. IRC SP 114 (2018) — *Guidelines for Seismic Design of Road Bridges.*
7. IS 800 (2007) — *Indian Standard: General Construction in Steel — Code of Practice.*
8. IS 2062 (2011) — *Hot Rolled Medium and High Tensile Structural Steel — Specification.*
9. Subramanian, N. (2008). *Design of Steel Structures.* Oxford University Press.
10. Steel-INSDAG Teaching Resource Materials. `https://www.steel-insdag.org`